

6(a)	$E = V / d$ $d = 350 / 1.4 \times 10^4$	C1
	$= 0.025 \text{ m}$	A1
6(b)(i)	$E = F / Q$	C1
	$Q = 6.7 \times 10^{-15} / 1.4 \times 10^4 (= 4.8 \times 10^{-19} \text{ C})$ $= (4.8 \times 10^{-19} / 1.6 \times 10^{-19}) e$	C1
	$= 3.0 e$	A1
6(b)(ii)	mass $= 8.3 \times 10^{-27} / 1.66 \times 10^{-27}$ $= 5.0 \text{ u}$	A1
6(b)(iii)	number $= 5 - 3$ $= 2$	A1